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A Discrete 3d−4f Metallacage as an Efficient Catalytic Nanoreactor for a Three-Component Aza-Darzens Reaction

Shengbin Zhou, Zhichao Zhang, Dongjie Bai, Jingzhe Li, Xiang Cui, Zhichuan J. Xu, Yu Tang, Xiaoliang Tang,* and Weisheng Liu

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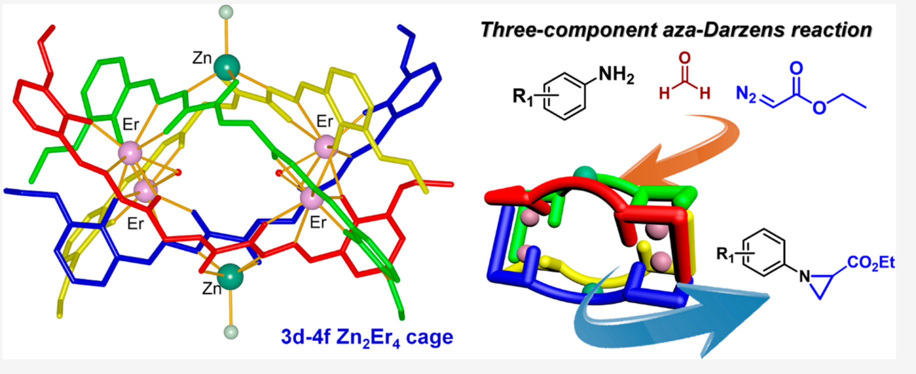

ABSTRACT: The exploration and development of coordination nanocages can provide an approach to control chemical reactions beyond the bounds of the flask, which has aroused great interest due to their significant applications in the field of molecular recognition, supramolecular catalysis, and molecular self-assembly. Herein, we take the advantage of a semirigid and nonsymmetric bridging ligand (H_5**L**) with rich metal-chelating sites to construct an unusual and discrete 3d−4f metallacage, $[Zn_2Er_4(H_2L)_4(NO_3)Cl_2(H_2O)]\cdot NO_3\cdot xCH_3OH\cdot yH_2O$ (**Zn_2Er_4**). The 3d−4f **Zn_2Er_4** cage possesses a quadruple-stranded structure, and all of the ligands wrap around an open spherical cavity within the core. The self-assembly of the unique cage not only ensures the structural stability of the **Zn_2Er_4** cage as a nanoreactor in solution but also makes the bimetallic lanthanide cluster units active sites that are exposed in the medium-sized cavity. It is important to note that the **Zn_2Er_4** cage as a homogeneous catalyst has been successfully applied to catalyze three-component aza-Darzens reactions of formaldehyde, anilines, and α-diazo esters without another additive under mild conditions, displaying better catalytic activity, higher specificity, short reaction time, and low catalyst loadings. A possible mechanism for this three-component aza-Darzens reaction catalyzed by the **Zn_2Er_4** cage has been proposed. These experimental results have demonstrated the great potential of the discrete 3d−4f metallacage as a host nanoreactor for the development of supramolecular or molecular catalysis.

■ INTRODUCTION

The coordination-driven self-assembly of polymetallic lanthanide complexes, including lanthanide helicates,[1] lanthanide molecular cages,[2] lanthanide coordination polymers,[3] lanthanide clusters,[4] and lanthanide metal−organic frameworks,[5] has received fairly extensive attention in the field of chemistry and materials science.[6] These polymetallic complexes not only often present unexpected structural aesthetics due to their inherent f orbital nature and high coordination numbers of lanthanide ions but also can provide a great number of catalytic active centers or subtle interactions between two or more metal ions, endowing them with an irreplaceable function and great potential applications in sensors, catalysis, magnetics, and so on.[7−10] In particular, coordination nanocages constructed on polymetallic lanthanide assemblies or container molecules have a unique environment within their cores, which can recognize guest molecules according to the size of the internal space and stabilize the transition state of molecules.[11] Hence, the design and synthesis of lanthanide coordination nanocages provide a delicate approach to control chemical reactions beyond the bounds of the flask.[12]

Many high-nuclearity coordination nanocages, such as common tetrahedral and octahedral cages, have been regulated by using the same transition-metal ions or lanthanide ions.[13−15] However, the controllable self-assembly and application of 3d−4f metallacages remain some of the great

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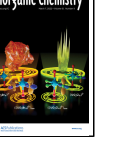

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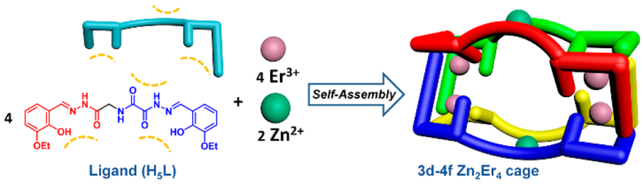
challenges due to the uncontrollability of the polynuclear arrangement and the difference of metal ion species in coordination number, coordination atom type, coordination mode, and stereochemical preference.[16] When d-block transition-metal ions and f-block lanthanide ions actively work together to construct a supramolecular cage, more metal nodes or second building units will be combined into a single integrated structure to generate potentially new synergistic effects, which will provide more possibilities for enriching the type of spatial structure and function of the fascinating cage to a great extent. As a reaction vessel on the nanoscale, a 3d−4f metallacage coated by some bridging ligands and two different kinds of metal ions may facilitate the reactions of substrates in a suitable cavity or on matching active sites and exhibit good solubility and high catalytic activity to promote reaction rates similarly to enzymes.[17] Therefore, it is valuable to explore and develop new 3d−4f metallacages and evaluate their catalytic functions.

On the other hand, the aza-Darzens reaction has been widely used in the synthesis of various biologically active agents with the smallest saturated aza-heterocycle, which is an useful precursor for other nitrogen-containing compounds via nucleophilic ring-opening reactions and other transformations.[18] Many small-molecule catalysts have been designed successfully and employed to synthesize substituted aziridines or chiral aziridines, which are versatile intermediates in antitumor and antibiotic drugs.[19] Although varieties of synthesis strategies are effective for the preparation of aziridines, most methods require complex starting materials or a multistep synthesis, and three-component aza-Darzens couplings have seldom been studied.[20] Raymond and Toste used for the first time a tetrahedral Ga(III) cage as a nanovessel catalyst to promote a three-component aza-Darzens reaction.[21] Lin et al. achieved multicomponent reactions of this type via the pores of Zr-MOF.[22] Significantly, these three-component aza-Darzens reactions based on different complex-based catalysts need a long reaction time, a high catalyst loading, or K_3PO_4 as an additive.

We have been interested in the construction of novel high-nuclearity clusters and the determination of the structure−activity relationship of these lanthanide complexes.[8a,17,23] To controllably prepare a supramolecular nanocage based on heterometallic complex and exploit the application of the coordination cage in catalysis, an unusual and discrete 3d−4f metallacage, $[Zn_2Er_4(H_2\mathbf{L})_4(NO_3)Cl_2(H_2O)]\cdot NO_3\cdot x CH_3OH\cdot y H_2O$ (**Zn_2Er_4**), was synthesized by the introduction of a semirigid and nonsymmetric bridging ligand ($H_5\mathbf{L}$) with rich metal-chelating sites, which could effectively fix more lanthanide ions and coordinate with transition-metal ions via the self-adjustment of different coordination atoms. The 3d−4f **Zn_2Er_4** cage possesses a quadruple-stranded structure, and all ligands wrap around open spherical space within the core (Scheme 1).

Lewis acidic sites on the lanthanide ions are exposed in the cavity of the **Zn_2Er_4** cage and a medium-sized cavity can act as a nanoreactor, simultaneously. It is important to note that the **Zn_2Er_4** cage as a homogeneous catalyst has been successfully applied to catalyze a three-component aza-Darzens reaction without another additive under mild conditions, displaying better catalytic activity, higher specificity, a short reaction time, and low catalyst loadings.

Scheme 1. Schematic Representation of the Synthesis of a 3d−4f Zn_2Er_4 Cage



■ EXPERIMENTAL SECTION

Materials and Instrumentation. All of the chemicals and solvents were commercially available and were used without further purification.

^{1}H NMR and ^{13}C NMR spectra were recorded with a JNM-ECS-400 MHz spectrometer, and the chemical shifts are reported in ppm by using tetramethylsilane as an internal reference. FT-IR spectra were recorded by using KBr pellets with a NEXUS 670 instrument in the region 4000−400 cm^{-1}. Elemental analyses (C, H, N) were performed with a Vario EL elemental analyzer. Thermogravimetric analysis (TGA) experiments were carried out with a PerkinElmer TG-7 analyzer heated from 25 to 800 °C at a heating rate of 10 °C/min under an Ar atmosphere. The X-ray powder diffraction (PXRD) data were recorded on a Rigaku D/Max-2500 diffractometer at 40 kV and 100 mA for a Cu-target tube and a graphite monochromator. HRESI-MS for the **Zn_2Er_4** cage was performed on Solarix XR with a Fourier transform ion cyclotron resonance mass spectrometer. The near-infrared spectrum was measured with an Edinburgh FLS920 combined steady-state transient fluorescence/phosphorescence spectrometer.

Synthesis. *Synthesis of Compound **a**.* In a 250 mL three-necked flask equipped with a stir bar and an addition funnel were placed glycine ethyl ester hydrochloride (3.071 g, 0.022 mol), triethylamine (9.0 mL, 0.065 mol), and anhydrous dichloromethane (100 mL). To this mixture was added a solution of oxalyl chloride monoethyl ester (2.73 g, 0.020 mol) in anhydrous dichloromethane (20 mL) dropwise under argon protection. The resulting solution was stirred at 0 °C for 10 h, and the reaction was quenched by the addition of water (50 mL). The mixture was washed with a saturated aqueous solution of sodium bicarbonate. The organic layer was dried with anhydrous sodium sulfate and concentrated using a rotary evaporator. A light yellow solid, ethyl 2-((2-ethoxy-2-oxoethyl)amino)-2-oxoacetate (**a**), was obtained (2.58 g, yield 63.5%). ^{1}H NMR (400 MHz, $CDCl_3$): δ 1.28 (t, J = 8.0 Hz, 3H), 1.38 (t, J = 8.0 Hz, 3H), 4.10 (d, J = 8.0 Hz, 2H), 4.24 (q, J = 8.0 Hz, 2H), 4.36 (q, J = 8.0 Hz, 2H), 7.53 (s, 1H). ^{13}C NMR (100 MHz, DMSO-d_6): δ 13.94, 14.09, 41.51, 61.78, 63.27, 156.82, 159.98, 168.79.

*Synthesis of Compound **b**.* Compound **a** (2.03 g, 0.01 mol) was added to a solution of 80% $NH_2NH_2\cdot H_2O$ (3.0 g, 0.06 mmol) in 40 mL of ethanol. The mixture was stirred for 10 h at reflux and then cooled to room temperature. The crude product was separated by filtration with cold ethanol to afford 2-hydrazinyl-*N*-(2-hydrazinyl-2-oxoethyl)-2-oxoacetamide (**b**) as a white solid (1.65 g, yield 94%). ^{1}H NMR (400 MHz, DMSO-d_6): δ 9.98 (s, 1H), 9.05 (s, 1H), 8.63 (t, J = 4.0 Hz, 1H), 4.49 (s, 2H), 4.16 (s, 2H), 3.68 (d, J = 4.0 Hz, 2H). ^{13}C NMR (100 MHz, DMSO-d_6): δ 167.89, 160.41, 158.21, 41.31.

*Synthesis of Ligand H_5**L**.* To a stirred solution of 3-ethoxy-2-hydroxybenzaldehyde (1.994 g, 0.012 mol) in 40 mL of ethanol was added **b** (0.876 g, 0.005 mol), and then the mixture was refluxed for 8 h at 78 °C. The white solid was filtered, further washed three times with cold methanol, and dried in air to afford H_5**L** as a white solid (1.96 g, yield 83%; the total yield is 49.5% based on the starting material oxalyl chloride monoethyl ester). ^{1}H NMR (400 MHz, DMSO-d_6): δ 1.32 (m, 6H), 3.92 (d, J = 8.0 Hz, 1H), 4.03 (m, 4H), 4.28 (d, J = 8.0 Hz, 1H), 6.79 (m, 2H), 6.98 (m, 2H), 7.07 (d, J = 8.0 Hz, 1.4H), 7.24 (dd, J = 4.0 and 8.0 Hz, 0.6H), 8.30−8.42 (d, J = 40 Hz, 1H), 8.76 (d, J = 4.0 Hz, 1H), 8.92 (t, J = 8.0 Hz, 0.6H), 9.18 (s, 1H), 10.66 (m, 1.4H), 11.51−11.56 (d, J = 40 Hz, 1H), 12.47 (d, J =

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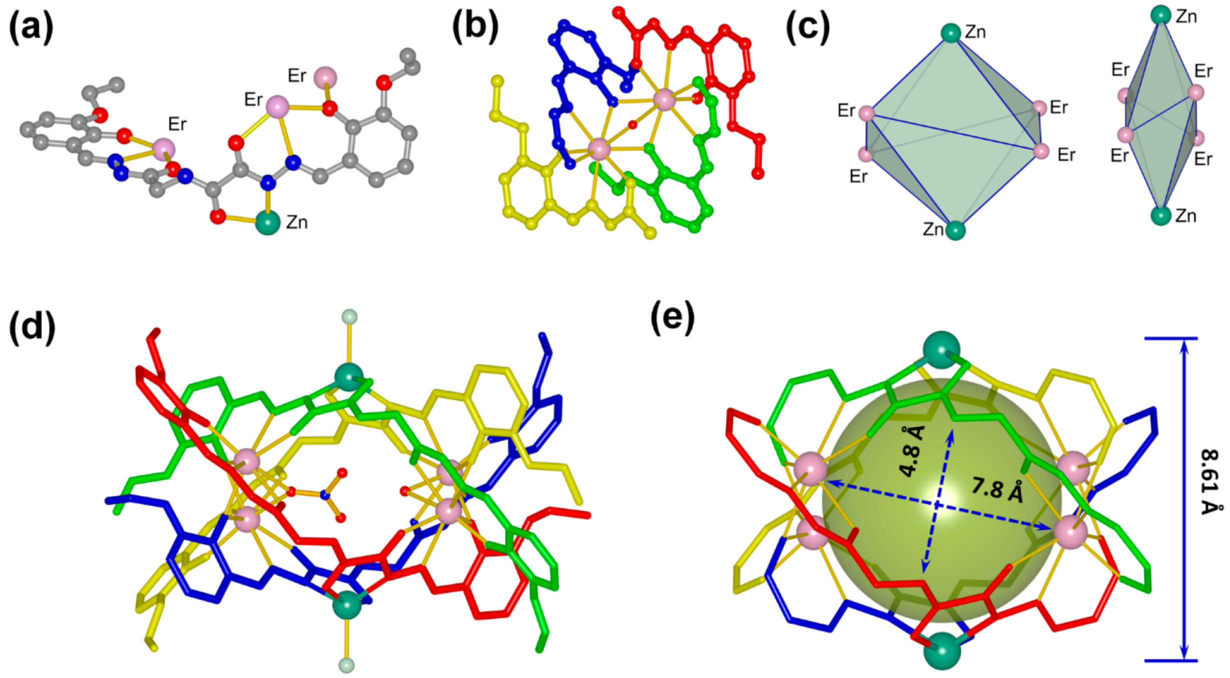

Figure 1. (a) Capped-stick representation and coordination mode of each deprotonated H_2L^{3-}. (b) Lanthanide-based bimetallic cluster unit connected by two deprotonated phenol groups and an oxygen atom. (c) Spatial arrangement and constructed polyhedra of all metal ions in different directions. (d) Molecular structure of the $\mathbf{Zn_2Er_4}$ cage with a quadruple strand. (e) Cavity and open windows of the $\mathbf{Zn_2Er_4}$ cage. Each ligand is represented in a different color. All hydrogen atoms, solvent molecules, and uncoordinated anions are omitted for clarity. Color code for (a): C, gray; O, red; N, blue; Cl, pale green; Er, pink; Zn, turquoise.

8.0 Hz, 1H). ^{13}C NMR (100 MHz, DMSO-d_6): δ 169.25, 164.89, 160.55, 160.17, 156.62, 156.54, 151.68, 147.97, 147.57, 146.68, 141.97, 121.58, 121.35, 121.10, 120.90, 119.65, 119.41, 119.29, 118.33, 118.14, 64.59, 15.17. ESI-MS m/z: calcd for $C_{22}H_{25}N_5O_7$ 471.1754, found 472.1835 $[M + H]^+$.

Synthesis of the Zn_2Er_4 Cage. A mixture of $H_5\mathbf{L}$ (23.6 mg, 0.05 mmol) and triethylamine 16 μL (0.22 mmol) in methanol (2 mL) was stirred for 15 min to obtain a transparent yellow solution. Then, $ZnCl_2$ (6.8 mg, 0.05 mmol) dissolved in methanol (4 mL) was added dropwise and stirred at room temperature for 10 min to obtain a suspension. $Er(NO_3)_3{\cdot}5H_2O$ (22.2 mg, 0.05 mmol) in methanol (2 mL) was added to this mixed system, and a clear solution was obtained. After 1 week, yellow lump crystals suitable for crystal analysis were filtered, washed three times with cold methanol, and dried in air (yield 49.0%; the total yield is 24.3% based on starting material oxalyl chloride monoethyl ester). ESI-MS $\{[Er_4Zn_2(H_2\mathbf{L})_4Cl_2]^{2+}{\cdot}NO_3^-{\cdot}4H_2O{\cdot}2CH_3OH+H^+\}^{2+}$ m/z: calcd. for $[C_{90}H_{105}N_{21}O_{37}Cl_2Zn_2Er_4]^{2+}$ 1472.6243, found 1472.6239. IR (KBr pellets, cm^{-1}): 3594.9(m), 3528.1(m), 3293.3(m), 3025.9(m), 3051.6(m), 2997.9(m), 2917.9(m), 2866.0(m), 1670.1(s), 1582.7(s), 1545.0(s), 1450.7(s), 1377.0(s), 1325.6(s), 1281.0(m), 1251.8(s), 1207.3(s), 1171.3(w), 1119.8(m), 1090.7(m), 1068.5(m), 1054.8(m), 1023.9(m), 981.1(m), 891.9(m), 849.1(m), 804.5(m), 737.7(m), 663.9(m), 569.7(m), 502.8(m), 451.4(m).

Single-Crystal X-ray Diffraction Analysis. Single-crystal X-ray diffraction measurements of the $\mathbf{Zn_2Er_4}$ cage were carried out on a Bruker SMART APEX-II CCD diffractometer with graphite-monochromated Mo Kα (λ = 0.71073 Å) radiation. A multiscan absorption correction was applied with the SADABS program. Unit cell dimensions were obtained with least-squares refinements, and all structures were solved by direct methods using SHELXS-2017.[24] Metal atoms in each complex were located from E maps. The non-hydrogen atoms were located in successive difference Fourier syntheses. The final refinement was performed by full-matrix least-squares methods with anisotropic thermal parameters for non-hydrogen atoms on F^2. The hydrogen atoms were introduced at calculated positions and not refined (riding model). Crystallographic data as well as details of the data collection and refinement for the $\mathbf{Zn_2Er_4}$ cage are summarized in Table S1 in the Supporting Information, whereas selected bond lengths and angles are given in Table S2 in the Supporting Information. The crystal data file in CIF format was deposited with the CCDC with the reference number 2113865, and these data can be obtained from the Cambridge Crystallographic Data Center.

■ RESULTS AND DISCUSSION

Design and Synthesis. Lanthanide ions with special electronic configurations exhibit high coordination numbers, kinetic lability, weak stereochemical preference, and variable coordination spheres. These properties generally result in uncontrollable and unpredictable stereostructures of lanthanide complexes. To effectively fix lanthanide ions and reduce the uncertainty of the coordination mode, it is necessary to design and select multidentate ligands. We have found that acylhydrazone groups linked together with a phenol moiety at the terminal of a ligand with multiple N and O donor atoms could provide stable tridentate-chelating site to capture a lanthanide ion, which is a very useful method to construct lanthanide-based nodes. Moreover, the existence of a hydroxyl oxygen atom could bridge other lanthanide ions to form a polynuclear arrangement. For a linker ligand, a rigid structure would be very advantageous for the self-assembly of a coordination cage.

On the basis of a consideration of the electrophilic characteristics of lanthanide ions and the difference in metal ion species, a semirigid and nonsymmetric ligand ($H_5\mathbf{L}$) was first designed and prepared by the introduction of non-symmetric dihydrazide to form dihydrazone with a dentate-bridging metal ion function (Figures S1−S7 in the Supporting Information). The ligand not only provides two stable tridentate-chelating sites for a lanthanide ion at the two

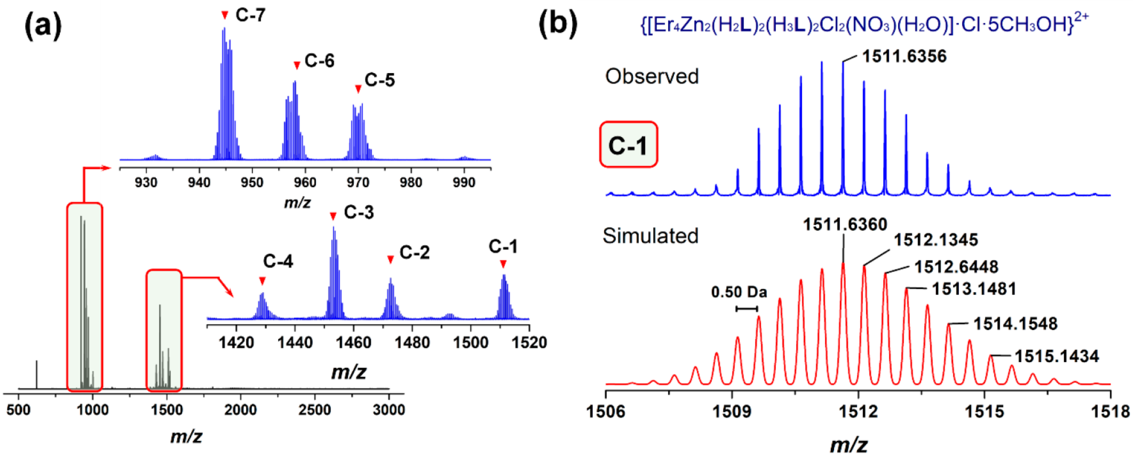

Figure 2. (a) HRESI mass spectra of the Zn_2Er_4 cage in methanol. Inset: magnified mass spectra in the ranges from *m*/*z* 920 to 1000 and from *m*/*z* 1430 to 1520. (b) Observed and theoretical mass peaks of **C-1** at *m*/*z* 1511.6356 with isotopic distribution patterns separated by 0.50 ± 0.005 Da, in good agreement with the positively doubly charged component $\{[Er_4Zn_2(H_2L)_4Cl_2]^{2+}\cdot NO_3^{-}\cdot Cl^{-}\cdot H_2O\cdot 5CH_3OH + 2H^{+}\}^{2+}$.

termini but also possesses rich metal-chelating sites in the middle to coordinate with a transition metal ion via a self-adjustment. Thus, the discrete 3d−4f metallacage $\mathbf{Zn_2Er_4}$ was readily synthesized by the addition of $Er(NO_3)_3\cdot 5H_2O$ and $ZnCl_2$ to a solution of $H_5\mathbf{L}$ and triethylamine in methanol. Yellow lump crystals suitable for X-ray diffraction analysis were obtained by slow evaporation of the solution at room temperature. FT-IR spectra, thermogravimetric analyses (TGA), and powder X-ray diffraction (PXRD) data of the $\mathbf{Zn_2Er_4}$ cage were investigated, and the results clearly indicated the formation of a 3d−4f complex (Figures S8−S10 in the Supporting Information).

Structure Description. A view of the X-ray crystal structure of the $\mathbf{Zn_2Er_4}$ cage is shown in Figure 1, and the cage crystallized in the tetragonal system with the non-centrosymmetric $P\bar{4}n2$ space group (No. 118).

Every ligand can coordinate with three lanthanide Er^{3+} ions and one Zn^{2+} ion through a self-adjustment of the ligand skeleton (Figure 1a). The cage mainly consists of four deprotonated $H_2\mathbf{L}^{3-}$ ligands, four Er^{3+} ions, and two Zn^{2+} ions. Each of the Er^{3+} ions is first chelated by two tridentate coordination moieties, both of which come from different ligands and are comprised of an acylhydrazone moiety and a phenol group on one end of the $H_2\mathbf{L}^{3-}$ ligand. Then, two coordinated Er^{3+} units are further connected to each other to form a lanthanide-based bimetallic cluster unit (the Er···Er distance is 3.54 Å) via one of the other's deprotonated phenol groups (Figure 1b) and two pairs of bimetallic cluster units are assembled by four ligands to form a quadruple-stranded helicate, simultaneously. Meanwhile, the Zn^{2+} ion is pentacoordinated by two enols of amides from different ligands and a Cl^{-} anion (Figure S11 in the Supporting Information), and two Zn^{2+} ions are fixed effectively at the equator of the helicate in opposite positions (the Zn···Zn distance is 8.61 Å). The polyhedron composed of six metal ions can be described as a distorted-octahedral structure (Figure 1c). In the $\mathbf{Zn_2Er_4}$ cage, there is a spherical cavity with a diameter of about 7.5 Å (the shortest distance between C12 and C13 in the opposite positions), in which one coordinated μ_2-NO_3− and one μ_2-bridging water molecule occupy disordered residual coordination sites of Er^{3+} ions (Figure 1d), and the octacoordinated polyhedra around each lanthanide ion finally can be described as slightly distorted dicapped trigonal prisms (Figure S12 in the Supporting Information). In addition, as shown in Figure 1e, two elliptical open windows (the minor axis is 4.8 Å and the long axis is 7.8 Å) are exposed outside the surface of the cavity in opposite positions because of unoccupied coordination sites, which may be beneficial to the entrance and exit of guest molecules in the nanocage.

Stability of the Zn_2Er_4 Cage in Solution. The stability of the $\mathbf{Zn_2Er_4}$ cage in solution was confirmed further by high-resolution electrospray ionization mass spectra (HRESI-MS). As shown in Figure 2, the presence of the intense peaks at *m*/*z* 1511.6356, 1472.6243, 1453.5944, and 1429.1022 with an isotopic distribution pattern separated by 0.50 ± 0.005 Da could belong to the positively doubly charged derivative ions of the $\mathbf{Zn_2Er_4}$ $[Er_4Zn_2(H_2\mathbf{L})_4Cl_2]$ cage $\{[Er_4Zn_2(H_2\mathbf{L})_4Cl_2]^{2+}\cdot NO_3^{-}\cdot Cl^{-}\cdot H_2O\cdot 5CH_3OH + 2H^{+}\}^{2+}$ (**C-1**, calcd *m*/*z* 1511.6360), $\{[Er_4Zn_2(H_2\mathbf{L})_4Cl_2]^{2+}\cdot NO_3^{-}\cdot 4H_2O\cdot 2CH_3OH + H^{+}\}^{2+}$ (**C-2**, calcd *m*/*z* 1472.6239), $\{[Er_4Zn_2(H_2\mathbf{L})_4Cl_2]^{2+}\cdot 2NO_3^{-}\cdot H_2O + H^{+}+NH_4^{+}\}^{2+}$ (**C-3**, calcd *m*/*z* 1453.5944), and $\{[Er_4Zn_2(H_2\mathbf{L})_4Cl_2]^{2+}\cdot CO_3^{2-}\cdot H_2O\cdot CH_3OH + 2H^{+}\}^{2+}$ (**C-4**, calcd *m*/*z* 1429.1020), respectively. In these positively doubly charged derivative ions, H^{+} and NH_4^{+} are common ion sources. However, a carbonate counteranion was found in **C-4**. The carbonate may come from the conversion of carbon dioxide in the air during the crystal growth process of the $\mathbf{Zn_2Er_4}$ cage. Meanwhile, the peaks at *m*/*z* 970.4146, 958.0829, and 945.4117 with an isotopic distribution pattern separated by 0.33 ± 0.005 Da also could be attributed to the tripositive derivative ions of $\mathbf{Zn_2Er_4}$ cage, $\{[Er_4Zn_2(H_2\mathbf{L})_4Cl_2]^{2+}\cdot NO_3^{-}\cdot 2H_2O\cdot 2CH_3OH + 2H^{+}\}^{3+}$ (**C-5**, calcd *m*/*z* 970.4147), $\{[Er_4Zn_2(H_2\mathbf{L})_4Cl_2]^{2+}\cdot 7H_2O + H^{+}\}^{3+}$ (**C-6**, calcd *m*/*z* 958.0832), and $\{[Er_4Zn_2(H_2\mathbf{L})_4Cl_2]^{2+}\cdot Cl^{-}\cdot 2H_2O + 2NH_4^{+}\}^{3+}$ (**C-7**, calcd *m*/*z* 945.4112), respectively. These fragments of the $\mathbf{Zn_2Er_4}$ cage have been analyzed in detail, and experimental HRESI-MS data are all in good agreement with the theoretical isotopic distribution and provide direct evidence for the formation and presence of stable 3d−4f metallacages in solution (Figures S13−S18 and Table S3 in the Supporting Information).

Optimization of Reaction Conditions. Such a unique and stable cage structure stimulated us to explore the catalytic

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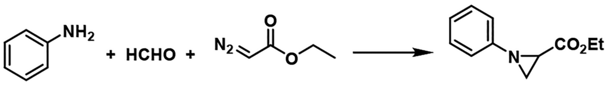
application of the Zn_2Er_4 cage. Fortunately, we found that the 3d−4f Zn_2Er_4 cage can serve as a kind of homogeneous catalyst for the synthesis of aziridine derivatives. The optimization of the reaction conditions of the three-component aza-Darzens reaction using aniline, formaldehyde, and ethyl diazoacetate as model substrates is summarized in Table 1.

Table 1. Optimization of Reaction Conditions for Three-Component Aza-Darzens Reaction



4a

entry[a]	catalyst	catalyst amt (mol %)	solvent	reaction time (h)	yield (%)[b]
1	$\mathbf{Zn_2Er_4}$	0.5	C_2H_5OH	2	49
2	$\mathbf{Zn_2Er_4}$	0.5	CH_3OH	2	58
3	$\mathbf{Zn_2Er_4}$	0.5	CH_2Cl_2	2	17
4	$\mathbf{Zn_2Er_4}$	0.5	1,4-dioxane	2	21
5	$\mathbf{Zn_2Er_4}$	0.5	CH_3CN	2	17
6	$\mathbf{Zn_2Er_4}$	0.5	DMF	2	37
7	$\mathbf{Zn_2Er_4}$	0.5	THF	2	14
8	$\mathbf{Zn_2Er_4}$	0.6	CH_3OH	2	66
9	$\mathbf{Zn_2Er_4}$	0.7	C_2H_5OH	2	72
10	$\mathbf{Zn_2Er_4}$	0.7	CH_3OH	2	73
11	$\mathbf{Zn_2Er_4}$	0.8	CH_3OH	2	74
12	$\mathbf{Zn_2Er_4}$	0.8	CH_3OH	3	83
13	$\mathbf{Zn_2Er_4}$	0.8	CH_3OH	4	84
14	$\mathbf{Zn_2Er_4}$	0.8	CH_3OH	16	86
15	H_5L	0.8	CH_3OH	2	trace
16	$ZnCl_2$	0.8	CH_3OH	2	trace
17	$Er(NO_3)_3{\cdot}5H_2O$	0.8	CH_3OH	2	7
18	$ZnCl_2$ + $Er(NO_3)_3{\cdot}5H_2O$	0.8 + 0.8[c]	CH_3OH	2	9
19	H_5L + $ZnCl_2$	0.8 + 0.8[d]	CH_3OH	3	trace
20	H_5L + $Er(NO_3)_3{\cdot}5H_2O$	0.8 + 0.8[e]	CH_3OH	3	13

[a]Reaction conditions unless specified otherwise: aniline 0.2 mmol, formaldehyde 0.4 mmol, α-diazo esters 0.4 mmol. [b]The yield was determined by ^1H NMR analysis by means of methoxybenzene/1,1,2,2-tetrachloroethane (0.2 mmol) as an internal standard. [c]$ZnCl_2$ (0.8 mol %) + $Er(NO_3)_3{\cdot}5H_2O$ (0.8 mol %). [d]$H_5\mathbf{L}$ (0.8 mol %) + $ZnCl_2$ (0.8 mol %). [e]$H_5\mathbf{L}$ (0.8 mol %) + $Er(NO_3)_3{\cdot}5H_2O$ (0.8 mol %).

In the presence of 0.5 mol % of the Zn_2Er_4 cage, the product **4a** of the three-component aza-Darzens reaction was obtained in 14−58% yields with different solvents, among which methanol as the best solvent exhibited an outstanding yield of above 58% due to its excellent solubility for all of the components including the catalyst Zn_2Er_4 cage. At the same time, through systematic screening of other parameters including catalyst loading and reaction time, the optimal reaction conditions of the Zn_2Er_4 cage for the three-component aza-Darzens reaction were obtained (0.8 mol % of Zn_2Er_4 cage in methanol at room temperature for 3 h), and the yield of the target product **4a** was 83% in the absence of any cocatalyst. In contrast, product **4a** was obtained in less than 13% yield in the presence of 0.8 mol % of $Er(NO_3)_3$, 1.6 mol % of a mixed salt (0.8 mol % of $ZnCl_2$ + 0.8 mol % of $Er(NO_3)_3{\cdot}5H_2O$), or a catalyst mixture (0.8 mol % of $H_5\mathbf{L}$ + 0.8 mol % of $Er(NO_3)_3{\cdot}5H_2O$) as the catalyst. In addition, **4a** could not be observed when the Zn_2Er_4 cage was replaced by the ligand $H_5\mathbf{L}$, $ZnCl_2$, or a catalyst mixture (0.8 mol % of $H_5\mathbf{L}$ + 0.8 mol % of $ZnCl_2$) as catalyst under identical conditions. These results indicated that the Zn_2Er_4 cage could effectively encapsulate small-molecule substrates into the host cage to provide a catalytic microenvironment in lanthanide metal positions for a three-component aza-Darzens reaction.

In addition, to further evaluate the stability of the Zn_2Er_4 cage during the reaction, the near-infrared (NIR) emission was utilized because of the presence of an Er^{3+} ion in the complex structure. The excitation and emission spectra of the catalyst Zn_2Er_4 cage in the solid state were recorded before and after the three-component aza-Darzens reaction. The Zn_2Er_4 cage exhibited obvious and broad characteristic emission of the Er^{3+} ion at 1540 nm, which can be assigned to a $^4I_{13/2} \rightarrow {}^4I_{15/2}$ transition, indicating efficient sensitization of Er^{3+} ion from ligands or the complex framework (Figure S19 in the Supporting Information). There was no change to the recycled catalyst Zn_2Er_4 cage in the excitation and NIR emission spectra, implying that structure of the Zn_2Er_4 cage was not damaged and the catalyst could exist stably in the reaction solution.

Universality of the Catalytic Reaction. Under the optimum experimental conditions, a series of substituted anilines were employed to investigate the generality and catalytic selectivity of the Zn_2Er_4 cage as a catalyst for a three-component aza-Darzens reaction, and the results are summarized in Table 2.

For electron-withdrawing substituents (p-NO_2) and halides (p-Cl, p-Br, p-I, o-I, and m-I) on the aniline ring, the reaction efficiency was low with about 30% yields (**4b−d**) or no reaction at all (**4e−g**). When electron-donating groups at the para position of aniline were employed, there were striking differences. The target product **4h** based on 4-methylaniline was obtained with a moderate yield of 57%, but 4-*tert*-butylaniline with a large steric hindrance group did not afford any of the desired aziridine. Significantly, when methoxy and ethoxy groups were located on the aniline ring, the corresponding products were obtained with up to 91% (**4i**) and 98% yields (**4j**), respectively (Figures S20−S44 in the Supporting Information). This might be due to the high reactivity of the condensation reaction between aqueous formaldehyde and aniline with an electron-donating effect in the cage. In addition, benzaldehyde, instead of formaldehyde, as one of the starting materials in three-component aza-Darzens reaction also did not afford any desired product, which might be because the aldehyde groups are not the main activated components in the Zn_2Er_4 cage and a large aromatic aldehyde affects the formation and migration of product molecules.

Mechanism of the Catalytic Reaction. On the basis of the above results, a possible mechanism for this three-component aza-Darzens reaction catalyzed by the Zn_2Er_4 cage was proposed. As shown in Figure 3a, the initial imine compound **I**, generated rapidly by the reaction of formaldehyde with aniline in solution, is followed by the encapsulation of the Zn_2Er_4 cage. In the cavity, a lanthanide Er^{3+} ion in a bimetallic cluster unit as a Lewis acid site coordinates with the N atom in imine **I** to form the complex intermediate **II**. Immediately, the α-diazo ester undergoes nucleophilic addition to the C atom of imine to form the

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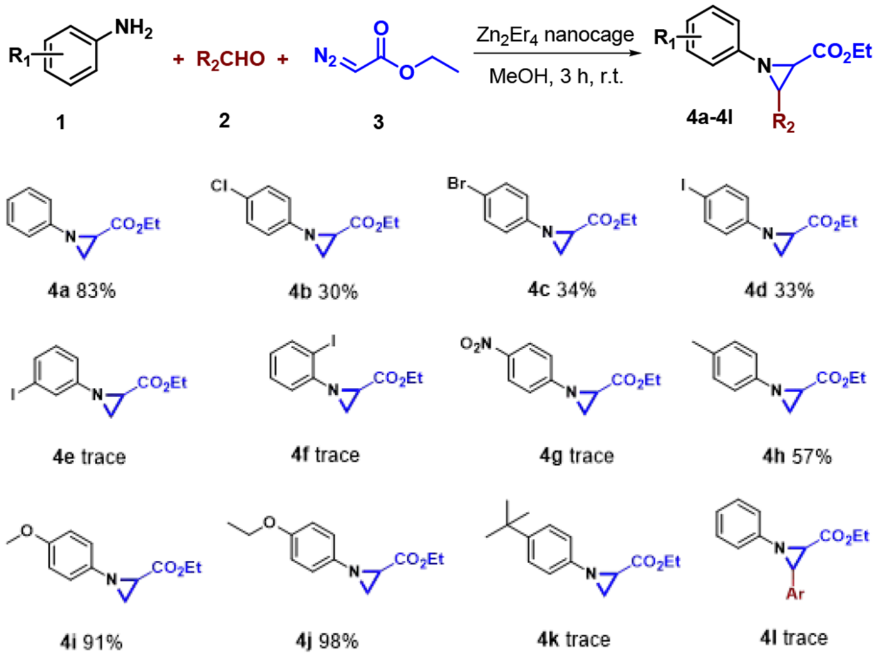
Table 2. Three-Component Aza-Darzens Reaction Catalyzed by the Zn_2Er_4 Cage as a Catalyst[a]



[a]Reaction conditions: substituted aniline 0.2 mmol, formaldehyde/benzaldehyde 0.4 mmol, α-diazo esters 0.4 mmol. The reaction temperature was room temperature. The yield was determined by means of ^{1}H NMR analysis with methoxybenzene/1,1,2,2-tetrachloroethane (0.2 mmol) as an internal standard.

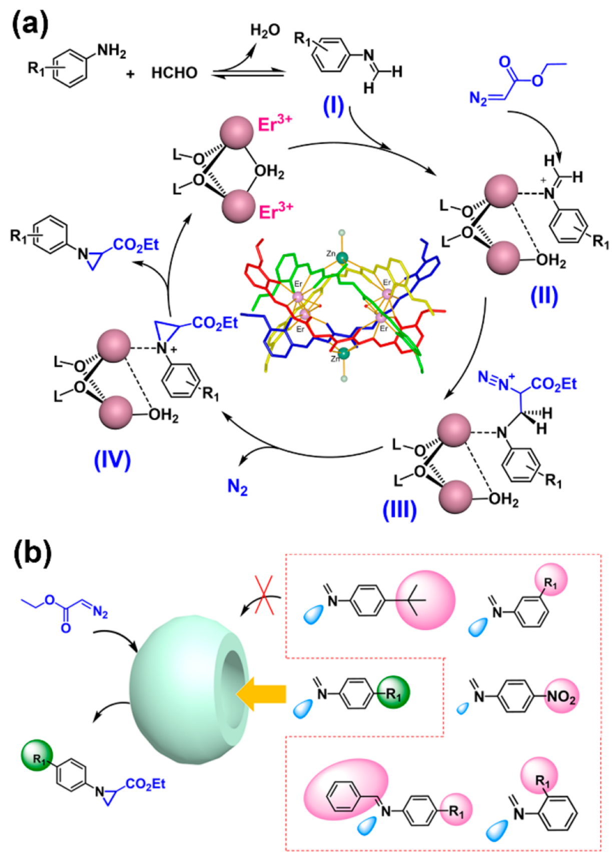


Figure 3. Proposed mechanism (a) and the possible reason for the selectivity of the reaction (b) for the three-component aza-Darzens reaction catalyzed by the Zn_2Er_4 cage.

branched intermediate **III**, which possesses such a relatively larger structure that the Er−N coordination bond is weakened and an intramolecular nucleophilic displacement of the N atom takes place simultaneously. The coordination ability of the intermediate product **IV** with an aza-heterocycle is further weakened, and the new molecule could be easily crowded out of the coordination cage. Finally, the target product aziridine with a chain structure is generated, accompanied by the release and regeneration of the catalyst Zn_2Er_4 cage. In addition, the better catalytic activity of the catalyst for the three-component aza-Darzens reaction of formaldehyde, anilines, and α-diazo esters but an inefficient reaction conversion for aromatic aldehydes and anilines with steric hindrance groups or electron-withdrawing substituents could be explained on the basis of the size of the Zn_2Er_4 cage (Figure 3b). The size of the open and small windows in the Zn_2Er_4 cage might play a key role in the selectivity of intermediates and the entrance or exit of guest molecules in the nanocage.

The interaction between the Zn_2Er_4 cage and guest molecules was explored by ^{1}H NMR, and the NMR signals of all H atoms in *N*-phenylmethanimine shifted by ∼0.2 ppm to higher magnetic fields there were no changes in the ^{1}H NMR signals of the host cage, which perhaps indicated indirectly that the imine intermediate *N*-phenylmethanimine as a guest could coordinate with the lanthanide ion centers in the cavity (Figure S45 in the Supporting Information).

■ CONCLUSION

In conclusion, we report herein the design and synthesis of an unusual and discrete 3d−4f metallacage constructed by four Er^{3+} ions, two Zn^{2+} ions, and four nonsymmetric bridging ligands with rich metal-chelating sites. The Zn_2Er_4 cage possesses a quadruple-stranded structure, and all of the ligands wrap around an open spherical cavity, in which the lanthanide ions are exposed and the medium-sized cavity can act as a nanoreactor. The Zn_2Er_4 cage as a homogeneous catalyst has been successfully applied to catalyze the three-component aza-Darzens reaction of formaldehyde, anilines, and α-diazo esters without another additive under mild conditions, different from the catalytic mechanism of classical catalysts used in the aza-Darzens reaction. The cage displays better catalytic activity and higher specificity with a shortened reaction time and a low catalyst loading. Moreover, the inefficient reaction conversion of the Zn_2Er_4 cage for aromatic aldehydes and anilines with steric hindrance groups or electron-withdrawing substituents provides evidence that the size of the host cavity and the rational arrangements of catalytic centers in the cage are crucial to the multicomponent reaction. To the best of our knowledge, the study represents an innovative and meaningful exploration for the design and application of a new discrete 3d−4f metallacage, which will create more opportunities to develop functional 3d−4f supramolecular cages as host nanoreactors.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.inorgchem.1c03729.

Characterization of the compounds, NMR, TGA, PXRD, HR-ESI, and additional figures and tables as described in the text (PDF)

Accession Codes

CCDC 2113865 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

■ AUTHOR INFORMATION

Corresponding Author

Xiaoliang Tang − *Key Laboratory of Nonferrous Metal Chemistry and Resources Utilization of Gansu Province and State Key Laboratory of Applied Organic Chemistry, Key Laboratory of Special Function Materials and Structure Design, Ministry of Education, College of Chemistry and Chemical Engineering, Lanzhou University, Lanzhou 730000, People's Republic of China*; orcid.org/0000-0001-9296-017X; Email: tangxiaol@lzu.edu.cn

Authors

Shengbin Zhou − *Key Laboratory of Nonferrous Metal Chemistry and Resources Utilization of Gansu Province and State Key Laboratory of Applied Organic Chemistry, Key Laboratory of Special Function Materials and Structure Design, Ministry of Education, College of Chemistry and Chemical Engineering, Lanzhou University, Lanzhou 730000, People's Republic of China*

Zhichao Zhang − *Key Laboratory of Nonferrous Metal Chemistry and Resources Utilization of Gansu Province and State Key Laboratory of Applied Organic Chemistry, Key Laboratory of Special Function Materials and Structure Design, Ministry of Education, College of Chemistry and Chemical Engineering, Lanzhou University, Lanzhou 730000, People's Republic of China*

Dongjie Bai − *Key Laboratory of Nonferrous Metal Chemistry and Resources Utilization of Gansu Province and State Key Laboratory of Applied Organic Chemistry, Key Laboratory of Special Function Materials and Structure Design, Ministry of Education, College of Chemistry and Chemical Engineering, Lanzhou University, Lanzhou 730000, People's Republic of China*

Jingzhe Li − *Key Laboratory of Nonferrous Metal Chemistry and Resources Utilization of Gansu Province and State Key Laboratory of Applied Organic Chemistry, Key Laboratory of Special Function Materials and Structure Design, Ministry of Education, College of Chemistry and Chemical Engineering, Lanzhou University, Lanzhou 730000, People's Republic of China*

Xiang Cui − *College of Chemistry and Chemical Engineering, Qinghai Normal University, Xining 810016, People's Republic of China*

Zhichuan J. Xu − *School of Materials Science and Engineering, Nanyang Technological University, Singapore 639798, Singapore*

Yu Tang − *Key Laboratory of Nonferrous Metal Chemistry and Resources Utilization of Gansu Province and State Key Laboratory of Applied Organic Chemistry, Key Laboratory of Special Function Materials and Structure Design, Ministry of Education, College of Chemistry and Chemical Engineering, Lanzhou University, Lanzhou 730000, People's Republic of China*; orcid.org/0000-0003-3933-043X

Weisheng Liu − *Key Laboratory of Nonferrous Metal Chemistry and Resources Utilization of Gansu Province and State Key Laboratory of Applied Organic Chemistry, Key Laboratory of Special Function Materials and Structure Design, Ministry of Education, College of Chemistry and Chemical Engineering, Lanzhou University, Lanzhou 730000, People's Republic of China*; orcid.org/0000-0001-5448-6315

Complete contact information is available at:
https://pubs.acs.org/10.1021/acs.inorgchem.1c03729

Notes

The authors declare no competing financial interest.

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